University of Illinois
ECE 310

Spring 2021
Profs. Moon, Katselis, and Shomorony

Midterm Exam 1

7:00-8:50 PM, Thursday, March 11, 2021

Please do not start the exam until the starting time.

- You may not use any books, electronic devices, or notes other than **one** handwritten two-sided sheet of 8.5" x 11" paper.
- You should solve the problems on blank sheets of paper, take pictures of your solutions, and upload them to Gradescope before the end of the exam time.

GOOD LUCK!

1. **(15 Pts.)** Answer **True** or **False** to each of the following statements:

   (a) An LTI system with transfer function $H(z) = \frac{1 - z^{-1}}{1 - 2z^{-1}}$ cannot be stable. **True**/[✓ **False**]

   (b) Let $X_1(z), X_2(z)$ be the rational z-transforms of $x_1[n], x_2[n]$. Then the poles of $X_1(z)$ and $X_2(z)$ must be poles of the z-transform of $x[n] = x_1[n] + x_2[n]$. Possible pole-zero cancellation **True**/[✓ **False**]

   (c) Cascade of two BIBO *unstable* LTI systems cannot be stable. **True**/[✓ **False**]

   (d) A causal LTI system with transfer function $H(z) = \frac{z^{-1}}{1 - z^{-1}}$ produces an unbounded output for input $x[n] = u[n]$. [✓ **True**]/**False**

   (e) Suppose $\sum_{n=-\infty}^{\infty} x[n]\delta[4\cos(2n\pi + \frac{\pi}{2}) - 6\sin(n\pi)] = 4$. Then, $\sum_{n=-\infty}^{\infty} x[n] = 3$. **True**/[✓ **False**]

2. **(10 Pts.)** Calculate the result of the following convolution: $\{-\underset{\uparrow}{1}, -2, 3, -3, 2, 1\} * \{-1, \underset{\uparrow}{0}, 1\}$.

   * See extra pages

3. **(24 Pts.)** For each of the systems with input $x[n]$ and output $y[n]$ in the table, indicate with YES or NO whether the properties indicated apply to the system.

	Linear	Time-Invariant	Causal	Stable
$y[n] = x[n]\cos(\frac{\pi}{3}(n-2))$	Yes	No	Yes	Yes
$y[n] = x[3]x[n]$	No	No	No	Yes
$y[n] = (\overbrace{\underbrace{0.8 + 0.8j}}^{\sqrt{1.28}\, e^{j\frac{\pi}{4}}})^n x[n]$	Yes	No	Yes	No

Goes to ∞ as $n \to \infty$.

4. **(10 Pts.)** Given the z-transform pair $x[n] \leftrightarrow X(z) = 1/(1 - 0.5z^{-1})$ with ROC: $|z| > 0.5$, determine the z-transform of $(n+1)x[n]$ and its ROC. * See extra pages

5. **(12 Pts.)** Suppose an LTI system has transfer function $H(z) = \frac{3z^{-1}}{1+z^{-2}}$ with ROC: $|z| > 1$.

   (a) **(8 Pts.)** The impulse response $h[n]$ of this system can be expressed as

$$h[n] = A\sin(\omega_0 n + \theta)u[n].$$

   * See extra pages

   Find the constants A, ω_0 and θ. **Hint:** One approach is to use the z-transform pair

$$\sin(\omega_0 n)u[n] \longleftrightarrow \frac{(\sin\omega_0)z^{-1}}{1 - 2(\cos\omega_0)z^{-1} + z^{-2}}, \quad |z| > 1.$$

   (b) **(4 Pts.)** Find a bounded real or complex-valued input signal $x[n]$ for which the corresponding output $y[n]$ is unbounded.

   * See extra pages

6. **(8 Pts.)** The output $y[n]$ and input $x[n]$ of a causal LTI system are related by the equation below. The system is initially at rest. Find the values of the impulse response $h[n]$ for the indicated values of n.

$y[n] = 0,\ n < 0$

$$y[n] = 2y[n-3] - x[n] + x[n-3]$$

$y[0] = 2y[-3] - \delta[0] + \delta[-3] = 1$

$y[1] = 2y[-2] - \delta[1] + \delta[-2] = 0$

$y[2] = 2y[-1] - \delta[2] + \delta[-1] = 0$

$y[3] = 2y[0] - \delta[3] + \delta[0] = 3$

$y[4] = 2y[1] - \delta[4] + \delta[1] = 0$

(a) $h[0] =$ 1

(b) $h[1] =$ 0

(c) $h[3] =$ 3

(d) $h[4] =$ 0

7. **(21 Pts.)** Suppose that the input to a causal LTI system is

$$x[n] = \frac{1}{4}\left(-\frac{1}{3}\right)^n u[n] - 3 \cdot 4^n u[-n-1]$$

↖RS ↖LS

and the z-transform of the output $y[n]$ is

$$Y(z) = \frac{13/4}{(1 - \frac{1}{2}z^{-1})(1 - z^{-1})(1 + \frac{1}{3}z^{-1})}.$$

(a) Find the z-transform of $x[n]$ and its ROC.

(b) Find the transfer function $H(z)$ and its ROC.

(c) Determine the ROC of $Y(z)$.

(d) Find the impulse response of the system.

(e) Is the system stable?

(a) $X(z) = \left(\frac{1}{4}\right)\left(\frac{1}{1+\frac{1}{3}z^{-1}}\right) + (3)\left(\frac{1}{1-4z^{-1}}\right)$, ROC: $\frac{1}{3} < |z| < 4$

$= \frac{\left(\frac{1}{4}\right)(1-4z^{-1}) + (3)(1+\frac{1}{3}z^{-1})}{(1+\frac{1}{3}z^{-1})(1-4z^{-1})}$

$X(z) = \frac{\frac{13}{4}}{(1+\frac{1}{3}z^{-1})(1-4z^{-1})}$

(b) $H(z) = \frac{Y(z)}{X(z)} = \frac{(1+\frac{1}{3}z^{-1})(1-4z^{-1})}{(1-\frac{1}{2}z^{-1})(1-z^{-1})(1+\frac{1}{3}z^{-1})}$

$H(z) = \frac{1-4z^{-1}}{(1-\frac{1}{2}z^{-1})(1-z^{-1})}$, $|z| > 1$ (causal)

(c) $R_Y =$ at least $R_x \cap R_h$ = at least $1 < |z| < 4$

$\frac{1}{3} < |z| < 4$ ↗ ↖ $|z| > 1$

↖ Pole @ $z = 4$ cancelled by $H(z)$

$R_Y = |z| > 1$

(d) $H(z) = \frac{A_1}{1-\frac{1}{2}z^{-1}} + \frac{A_2}{1-z^{-1}}$

$1 - 4z^{-1} = A_1(1-z^{-1}) + A_2(1-\frac{1}{2}z^{-1})$

A_1: $z = \frac{1}{2}$
$-7 = -A_1$
$A_1 = 7$

A_2: $z = 1$
$-3 = \frac{1}{2}A_2$
$A_2 = -6$

→ $h[n] = 7\left(\frac{1}{2}\right)^n u[n] - 6u[n]$

(e) Not stable, because ROC of $H(z)$ does not contain the unit circle.

② $\{\underset{\uparrow}{-1}, -2, 3, -3, 2, 1\} * \{-1, \underset{\uparrow}{0}, 1\} = y[n]$

$\quad\quad x[n] \quad\quad\quad\quad\quad\quad\quad h[n]$

$$y = Hx = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ -2 \\ 3 \\ -3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -4 \\ 1 \\ 1 \\ -4 \\ 2 \\ 1 \end{bmatrix} \begin{matrix} \\ \leftarrow n=0 \\ \\ \\ \\ \\ \\ \\ \end{matrix}$$

$y[n] = \{1, \underset{\uparrow}{2}, -4, 1, 1, -4, 2, 1\}$

④ $x[n] \overset{Z}{\longleftrightarrow} X(z) = \dfrac{1}{1-\frac{1}{2}z^{-1}}, \ |z| > \frac{1}{2}$

$y[n] = (n+1)x[n]$

$\quad = nx[n] + x[n]$

$Y(z) = -z\dfrac{dX(z)}{dz} + X(z)$

or use table pair for $na^n u[n]$

$\dfrac{dX(z)}{dz} = (-1)(1-\frac{1}{2}z^{-1})^{-2}(\frac{1}{2}z^{-2})$

$Y(z) = \dfrac{\frac{1}{2}z^{-1}}{(1-\frac{1}{2}z^{-1})^2} + \dfrac{1}{1-\frac{1}{2}z^{-1}}, \ |z| > \frac{1}{2}$

⑤ (a) * $H(z) = \dfrac{3z^{-1}}{1+z^{-2}}, \ |z| > 1$

* $\sin(w_0 n)u[n] \longleftrightarrow \dfrac{\sin(w_0)z^{-1}}{1-2\cos(w_0)+z^{-2}}, \ |z| > 1$

* $w_0 = \frac{\pi}{2}$: $\sin(\frac{\pi}{2}n)u[n] \longleftrightarrow \dfrac{z^{-1}}{1+z^{-2}}, \ |z| > 1$

$h[n] = 3\sin(\frac{\pi}{2}n)u[n] \longleftrightarrow H(z) = \dfrac{3z^{-1}}{1+z^{-2}}, \ |z| > 1$ ✓

$A = 3, \ w_0 = \frac{\pi}{2}, \ \theta = 0$

(b) * Poles @ $z = e^{\pm j\frac{\pi}{2}} = \pm j$

Let $x[n]$ any of:
- $j^n u[n]$
- $\cos(\frac{\pi}{2}n)u[n]$
- $\sin(\frac{\pi}{2}n)u[n]$